

Reflective Lantern - Innovation Report

New Project Built

Name: Cyber-Guard

URL: <https://github.com/atharvadevne123/reflective-lantern/pull/70>

Date: 2026-09-02

Inspired by: ML-for-security tooling (HN/GitHub trending scrape blocked by egress policy)

What it does: Real-time network intrusion detection and cybersecurity threat classification API using an XGBoost-RandomForest ensemble with KS-drift monitoring, IsolationForest anomaly detection, and automated retraining.

Measured Performance (5-fold stratified CV, n=600)

Accuracy: 0.960 +/- 0.021

AUC-ROC: 0.997 +/- 0.002 (weighted one-vs-rest)

All 5 threat signatures verified live at >0.98 confidence.

Required Stack Coverage: 100%

[X] Python

[X] FastAPI (5 endpoints)

[X] XGBoost+RandomForest ensemble

[X] sklearn Pipeline (15 features)

[X] Model monitoring (KS-drift)

[X] Docker + compose + PostgreSQL

[X] SQLAlchemy + Alembic

[X] pytest suite (102 tests)

[X] .env.example

[X] GitHub Actions CI

Optional Enhancements (6 of 4 required)

[X] Anomaly detection (IsolationForest)

[X] Automated retraining (Airflow DAG)

[X] Drift detection (KS-test)

[X] 5-fold CV + AUC-ROC

[X] MLflow experiment tracking

[X] AWS/boto3 S3 artifact stub

Project Stats

Files created: 31

Tests written: 102 (all passing)

Commits: 60

ruff check .: exits 0

Bugs Found and Fixed During the Build

1. Train/serve skew in rolling features: single-row inference produced rolling_src_std=0.00 vs training mean 426.66, pushing every request off the training manifold. The anomaly detector flagged 100% of traffic. Fixed by learning fallbacks at fit time; single-row and batch rates now agree exactly (3.00% vs 3.00%). Regression test confirmed to fail without fix.
2. Synthetic labels were drawn independently of features, capping AUC at chance (measured 0.490). Generator now samples conditional on threat class: AUC 0.490 -> 0.997.
3. metrics.json could contain bare NaN when a CV fold lacked a rare class (sklearn warns, does not raise). Now nanmean-averaged with allow_nan=False.
4. Test isolation: log_prediction commits, so fixture rollback was not enough.
5. scripts/seed_data.py could not import app when run directly.
6. /health reported healthy while the database was unreachable.

Note on Repository Location

This session's GitHub access was scoped to atharvadevne123/reflective-lantern. Both POST /user/repos and the GraphQL createRepository mutation were denied by the session proxy, so a standalone Cyber-Guard repository could not be created. The project was built as a self-contained subdirectory on branch cyber-guard-innovation and opened as draft PR #70; it can be extracted to its own repository unchanged. The HN and GitHub-trending scrape in Phase A was also blocked by the same egress policy, so the topic was selected against the duplicate-check gate alone (cyber/intrusion: zero overlap with 26 prior entries).